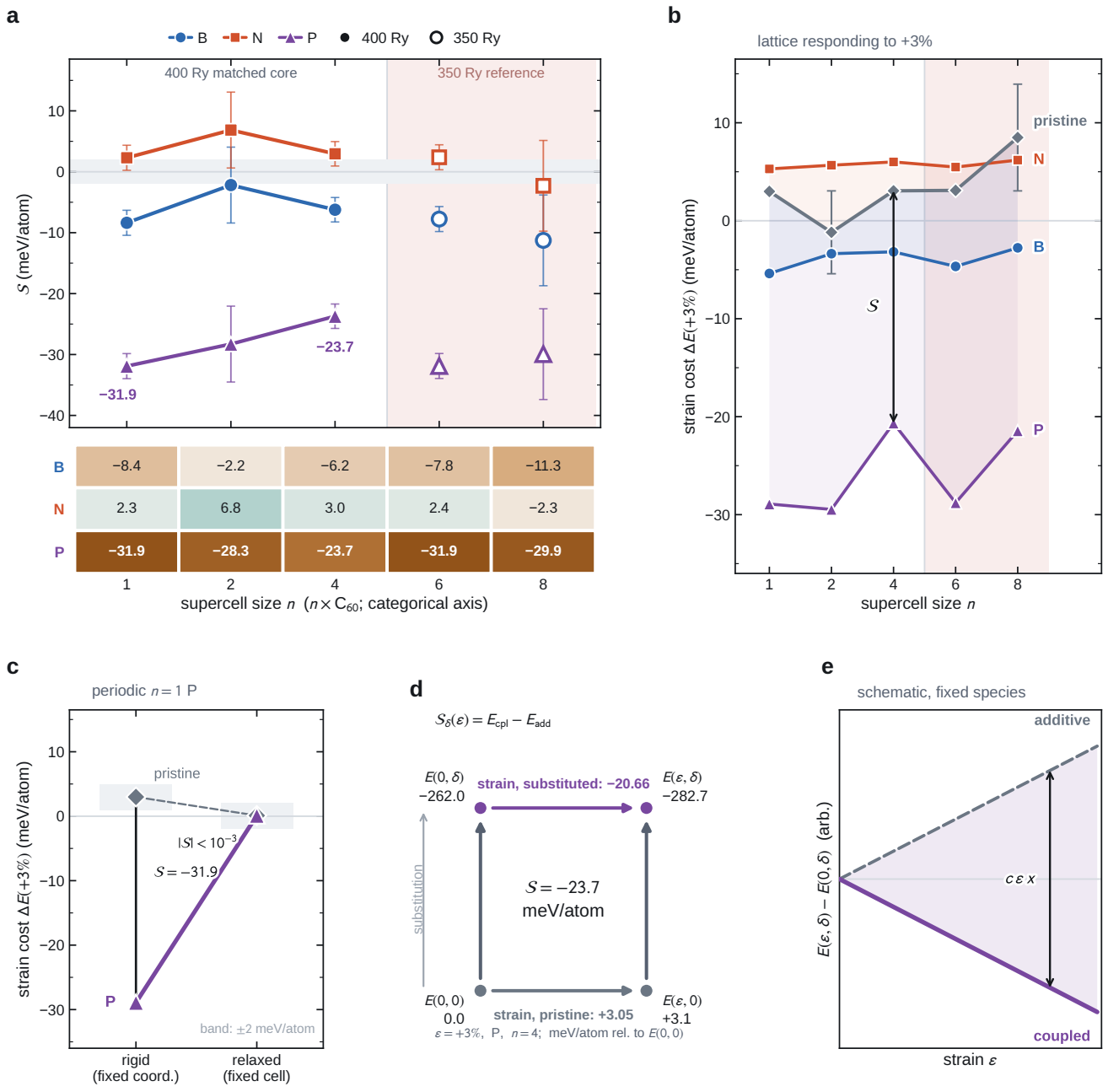

a
B
N
P
400 Ry
350 Ry
400 Ry matched core
350 Ry reference
S (meV/atom)
−23.7
−31.9
B −8.4 −2.2 −6.2 −7.8 −11.3
N 2.3 6.8 3.0 2.4 −2.3
P −31.9 −28.3 −23.7 −31.9 −29.9
supercell size n (n × C60; categorical axis)
b
lattice responding to +3%
pristine
N
B
P
S
strain cost ΔE(+3%) (meV/atom)
supercell size n
c
periodic n = 1 P
pristine
|S| < 10−3
S = −31.9
P
band: ±2 meV/atom
strain cost ΔE(+3%) (meV/atom)
rigid (fixed coord.)
relaxed (fixed cell)
d
Sδ(ε) = Ecpl − Eadd
E(0, δ) −262.0
strain, substituted: −20.66
E(ε, δ) −282.7
substitution
S = −23.7 meV/atom
E(0, 0) 0.0
strain, pristine: +3.05
E(ε, 0) +3.1
ε = +3%, P, n = 4; meV/atom rel. to E(0, 0)
e
schematic, fixed species
additive
coupled
c ε x
E(ε, δ) − E(0, δ) (arb.)
strain ε